

Math 161
Final Exam
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1. Let $A \subset B \subset \mathbb{R}$ with $A \neq \emptyset$. Prove that if B has a least upper bound, then so does A , and that in that case $\sup A \leq \sup B$.

2. Let $f, g : \mathbb{R} \rightarrow \mathbb{R}$ be functions satisfying the following conditions:

- $\lim_{x \rightarrow a} f(x) = L$;
- $x \neq a$ implies $f(x) \neq L$;
- $\lim_{y \rightarrow L} g(y)$ exists.

Prove that $\lim_{x \rightarrow a} g(f(x)) = \lim_{y \rightarrow L} g(y)$.

3. Suppose that $g : \mathbb{R} \rightarrow \mathbb{R}$ is continuous at 0, and let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) := xg(x)$. Prove that f is differentiable at 0, and express $f'(0)$ in terms of g .

4. Find the maximum and minimum values of the function

$$f(x) := \frac{x}{x^2 + 1}$$

on the interval $[0, 2]$. Justify your answer.

5. (Extra credit) Recall that a subset $A \subset \mathbb{R}$ is called *dense* if, for any $x, y \in \mathbb{R}$ such that $x < y$, there exists $z \in A$ such that $x < z < y$. Suppose that $f : \mathbb{R} \rightarrow \mathbb{R}$ is continuous and $A \subset \mathbb{R}$ is dense. Prove that if $f(x) = 0$ for all $x \in A$, then $f(x) = 0$ for all $x \in \mathbb{R}$. (Hint: Prove that $|f(x)| < \epsilon$ for any $\epsilon > 0$.)